

confocal_lacerate

```
library(tidyverse)
library(dplyr)
library(stringr)
library(rstatix)
library(patchwork)
library(car)
library(emmeans)
library(multcomp)
library(multcompView)
library(tibble)
```

```
rm(list = ls())
```

```
getwd()
```

```
## [1] "/Users/junbc/Documents/GitHub/heating_lacerates_final/code"
```

```
setwd("~/Documents/GitHub/heating_lacerates_final")
```

```
# Avoid conflicts with functions from other loaded packages
```

```
select    <- dplyr::select
filter    <- dplyr::filter
mutate    <- dplyr::mutate
summarise <- dplyr::summarise
summarize <- dplyr::summarize
group_by  <- dplyr::group_by
arrange   <- dplyr::arrange
rename    <- dplyr::rename
count     <- dplyr::count
left_join <- dplyr::left_join
```

```
# Data Processing
```

```
raw_confocal_data <- read_csv("data/Lacerate_EdU_Caspase.csv")
```

```
confocal_data <- raw_confocal_data %>%
```

```
  filter(True_Channel %in% c("EdU", "Caspase")) %>%
```

```
  mutate(
```

```
    marker = factor(True_Channel, levels = c("EdU", "Caspase")),
```

```
    time = factor(Time, levels = c("2dpl", "5dpl", "8dpl", "14dpl")),
```

```
    treatment = as.character(Treatment),
```

```
    state = case_when(
```

```
      str_detect(treatment, regex("Apo", ignore_case = TRUE)) ~ "Apo",
```

```
      str_detect(treatment, regex("Inoc", ignore_case = TRUE)) ~ "Inoc",
```

```
      str_detect(treatment, regex("Sym", ignore_case = TRUE)) ~ "Sym",
```

```
      TRUE ~ NA_character_
```

```

),
heat = case_when(
  str_detect(treatment, regex("HS|Heat", ignore_case = TRUE)) ~ "Heat",
  str_detect(treatment, regex("Control", ignore_case = TRUE)) ~ "Control",
  TRUE ~ NA_character_
),
state = factor(state, levels = c("Apo", "Inoc", "Sym")),
heat = factor(heat, levels = c("Control", "Heat")),
percentage = Percentage
) %>%
drop_na(marker, state, heat, time, percentage)

sample_size_table_wide <- confocal_data %>%
  count(marker, state, time, heat) %>%
  pivot_wider(names_from = heat, values_from = n)

print(sample_size_table_wide)

```

```

## # A tibble: 22 x 5
##   marker state time Control Heat
##   <fct> <fct> <fct>   <int> <int>
## 1 EdU    Apo   2dpl      3      6
## 2 EdU    Apo   5dpl      3      5
## 3 EdU    Apo   8dpl      5      4
## 4 EdU    Apo  14dpl      5      4
## 5 EdU    Inoc   5dpl      4      5
## 6 EdU    Inoc   8dpl      4      5
## 7 EdU    Inoc  14dpl      7      6
## 8 EdU    Sym    2dpl      5      6
## 9 EdU    Sym    5dpl     10      8
## 10 EdU   Sym    8dpl      6      8
## # i 12 more rows

```

```

p6_n <- confocal_data %>%
  count(marker, state, time, heat) %>%
  arrange(marker, state, time, heat)

print(p6_n, n = Inf)

```

```

## # A tibble: 44 x 5
##   marker state time heat      n
##   <fct> <fct> <fct> <fct>   <int>
## 1 EdU    Apo   2dpl Control    3
## 2 EdU    Apo   2dpl Heat       6
## 3 EdU    Apo   5dpl Control    3
## 4 EdU    Apo   5dpl Heat       5
## 5 EdU    Apo   8dpl Control    5
## 6 EdU    Apo   8dpl Heat       4
## 7 EdU    Apo  14dpl Control    5
## 8 EdU    Apo  14dpl Heat       4
## 9 EdU    Inoc   5dpl Control    4
## 10 EdU   Inoc   5dpl Heat       5

```

## 11	EdU	Inoc	8dpl	Control	4
## 12	EdU	Inoc	8dpl	Heat	5
## 13	EdU	Inoc	14dpl	Control	7
## 14	EdU	Inoc	14dpl	Heat	6
## 15	EdU	Sym	2dpl	Control	5
## 16	EdU	Sym	2dpl	Heat	6
## 17	EdU	Sym	5dpl	Control	10
## 18	EdU	Sym	5dpl	Heat	8
## 19	EdU	Sym	8dpl	Control	6
## 20	EdU	Sym	8dpl	Heat	8
## 21	EdU	Sym	14dpl	Control	5
## 22	EdU	Sym	14dpl	Heat	9
## 23	Caspase	Apo	2dpl	Control	3
## 24	Caspase	Apo	2dpl	Heat	6
## 25	Caspase	Apo	5dpl	Control	4
## 26	Caspase	Apo	5dpl	Heat	5
## 27	Caspase	Apo	8dpl	Control	6
## 28	Caspase	Apo	8dpl	Heat	4
## 29	Caspase	Apo	14dpl	Control	5
## 30	Caspase	Apo	14dpl	Heat	4
## 31	Caspase	Inoc	5dpl	Control	4
## 32	Caspase	Inoc	5dpl	Heat	6
## 33	Caspase	Inoc	8dpl	Control	4
## 34	Caspase	Inoc	8dpl	Heat	5
## 35	Caspase	Inoc	14dpl	Control	7
## 36	Caspase	Inoc	14dpl	Heat	5
## 37	Caspase	Sym	2dpl	Control	6
## 38	Caspase	Sym	2dpl	Heat	6
## 39	Caspase	Sym	5dpl	Control	10
## 40	Caspase	Sym	5dpl	Heat	8
## 41	Caspase	Sym	8dpl	Control	6
## 42	Caspase	Sym	8dpl	Heat	8
## 43	Caspase	Sym	14dpl	Control	5
## 44	Caspase	Sym	14dpl	Heat	9

```
ymax_edu <- confocal_data %>%
  filter(marker == "EdU") %>%
  summarise(max_val = max(percentage, na.rm = TRUE)) %>%
  pull(max_val)
```

```
ymax_cas <- confocal_data %>%
  filter(marker == "Caspase") %>%
  summarise(max_val = max(percentage, na.rm = TRUE)) %>%
  pull(max_val)
```

```
ymax_edu <- ymax_edu + 5
ymax_cas <- ymax_cas + 5
```

```
make_panel <- function(data_sub, panel_title, ylab = NULL, show_legend = FALSE, y_limit = NULL) {

  fit <- lm(percentage ~ heat * time, data = data_sub)

  emm <- emmeans(fit, ~ heat | time)
```

```

letters_df <- cld(emm, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group))

y_pos <- data_sub %>%
  group_by(time, heat) %>%
  summarise(y = max(percentage, na.rm = TRUE), .groups = "drop")

letters_df <- left_join(letters_df, y_pos, by = c("time", "heat")) %>%
  mutate(y = y + 6.5)

if (is.null(y_limit)) {
  y_max <- max(c(data_sub$percentage, letters_df$y), na.rm = TRUE) + 2
} else {
  y_max <- y_limit
}

ggplot(data_sub, aes(x = time, y = percentage, color = heat)) +
  geom_jitter(
    position = position_jitterdodge(jitter.width = 0.12, dodge.width = 0.75),
    size = 2.2,
    alpha = 0.9
  ) +
  geom_boxplot(
    aes(group = interaction(time, heat)),
    outlier.shape = NA,
    width = 0.6,
    fill = NA,
    linewidth = 0.8,
    position = position_dodge(width = 0.75)
  ) +
  geom_text(
    data = letters_df,
    aes(x = time, y = y, label = .group, group = heat),
    position = position_dodge(width = 0.75),
    inherit.aes = FALSE,
    fontface = "bold",
    size = 4,
    color = "black"
  ) +
  scale_color_manual(
    values = c("Control" = "blue", "Heat" = "red"),
    labels = c("Control" = "Control (25°C)", "Heat" = "Heat Stress (32°C)")
  ) +
  coord_cartesian(ylim = c(0, y_max), clip = "off") +
  labs(title = panel_title, x = NULL, y = ylab, color = NULL) +
  theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    legend.position = if (show_legend) "top" else "none",
    plot.margin = margin(8, 8, 8, 8)
  )
}

```

```

# Figure
# ---- EdU (shared scale)
p_edu_apo <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Apo"),
  "Cell proliferation - Apo",
  "EdU-positive area (%)",
  y_limit = ymax_edu
)

p_edu_inoc <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Inoc"),
  "Cell proliferation - Inoc",
  y_limit = ymax_edu
)

p_edu_sym <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Sym"),
  "Cell proliferation - Sym",
  show_legend = TRUE,
  y_limit = ymax_edu
)

# ---- Caspase (shared scale)
p_cas_apo <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Apo"),
  "Cell death - Apo",
  "Caspase-positive area (%)",
  y_limit = ymax_cas
)

p_cas_inoc <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Inoc"),
  "Cell death - Inoc",
  y_limit = ymax_cas
)

p_cas_sym <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Sym"),
  "Cell death - Sym",
  show_legend = TRUE,
  y_limit = ymax_cas
)

p_edu3 <- (p_edu_apo | p_edu_inoc | p_edu_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

p_cas3 <- (p_cas_apo | p_cas_inoc | p_cas_sym) +
  plot_layout(guides = "collect") &
  theme(legend.position = "top")

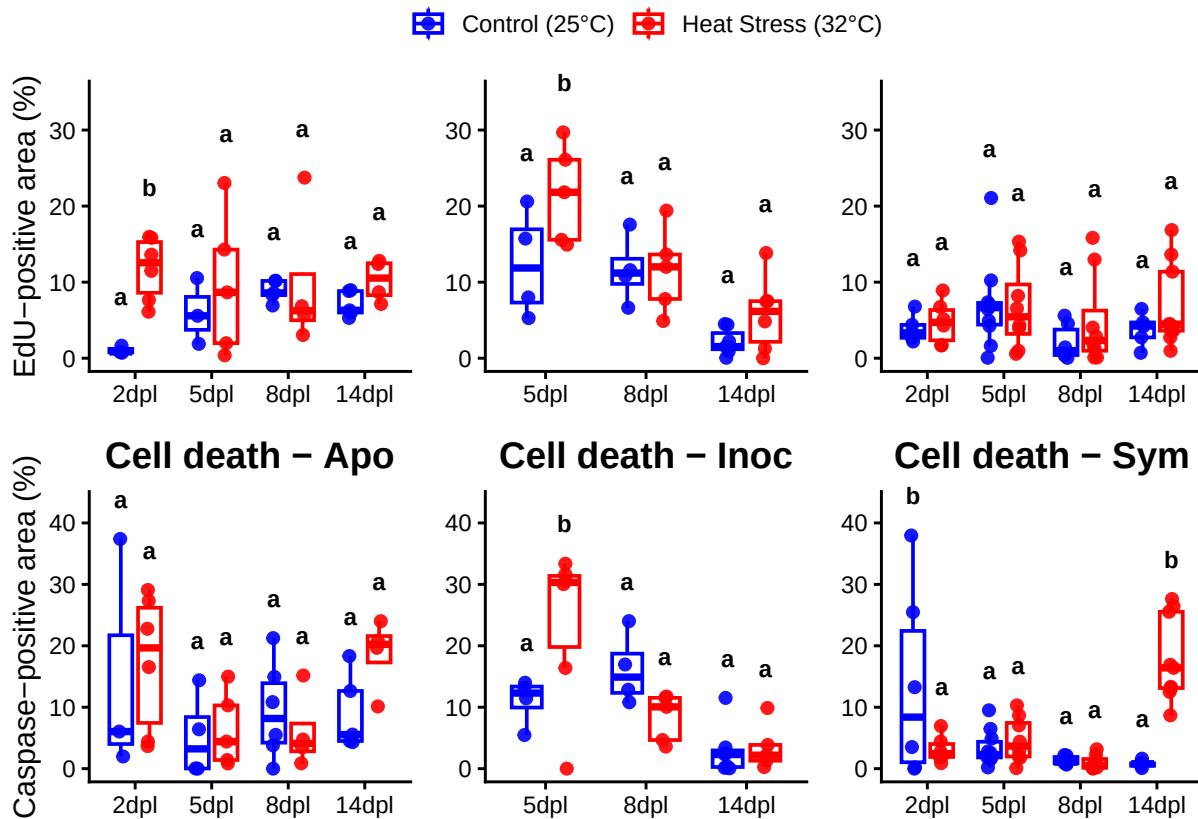
p6 <- (p_edu_apo | p_edu_inoc | p_edu_sym) /
  (p_cas_apo | p_cas_inoc | p_cas_sym) +

```

```
plot_layout(guides = "collect") &
theme(legend.position = "top")
```

```
print(p6)
```

Cell proliferation – ApoCell proliferation – InocCell proliferation – Syn



```
ggsave(
  filename = "Fig6_confocal.png",
  plot = p6,
  path = "figs",
  device = "png",
  width = 11,
  height = 8,
  units = "in",
  dpi = 600,
  bg = "white"
)
```

```
ggsave(
  filename = "Fig6_confocal.pdf",
  plot = p6,
  path = "figs",
  device = pdf,
  width = 11,
```

```

height = 8,
units = "in",
bg = "white"
)

# ---- EdU (shared scale)
p_edu_apo <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Apo"),
  "EdU - Apo",
  "EdU-positive area (%)",
  y_limit = ymax_edu
)

p_edu_inoc <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Inoc"),
  "EdU - Inoc",
  y_limit = ymax_edu
)

p_edu_sym <- make_panel(
  filter(confocal_data, marker == "EdU", state == "Sym"),
  "EdU - Sym",
  show_legend = TRUE,
  y_limit = ymax_edu
)

# ---- Caspase (shared scale)
p_cas_apo <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Apo"),
  "Caspase - Apo",
  "Caspase-positive area (%)",
  y_limit = ymax_cas
)

p_cas_inoc <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Inoc"),
  "Caspase - Inoc",
  y_limit = ymax_cas
)

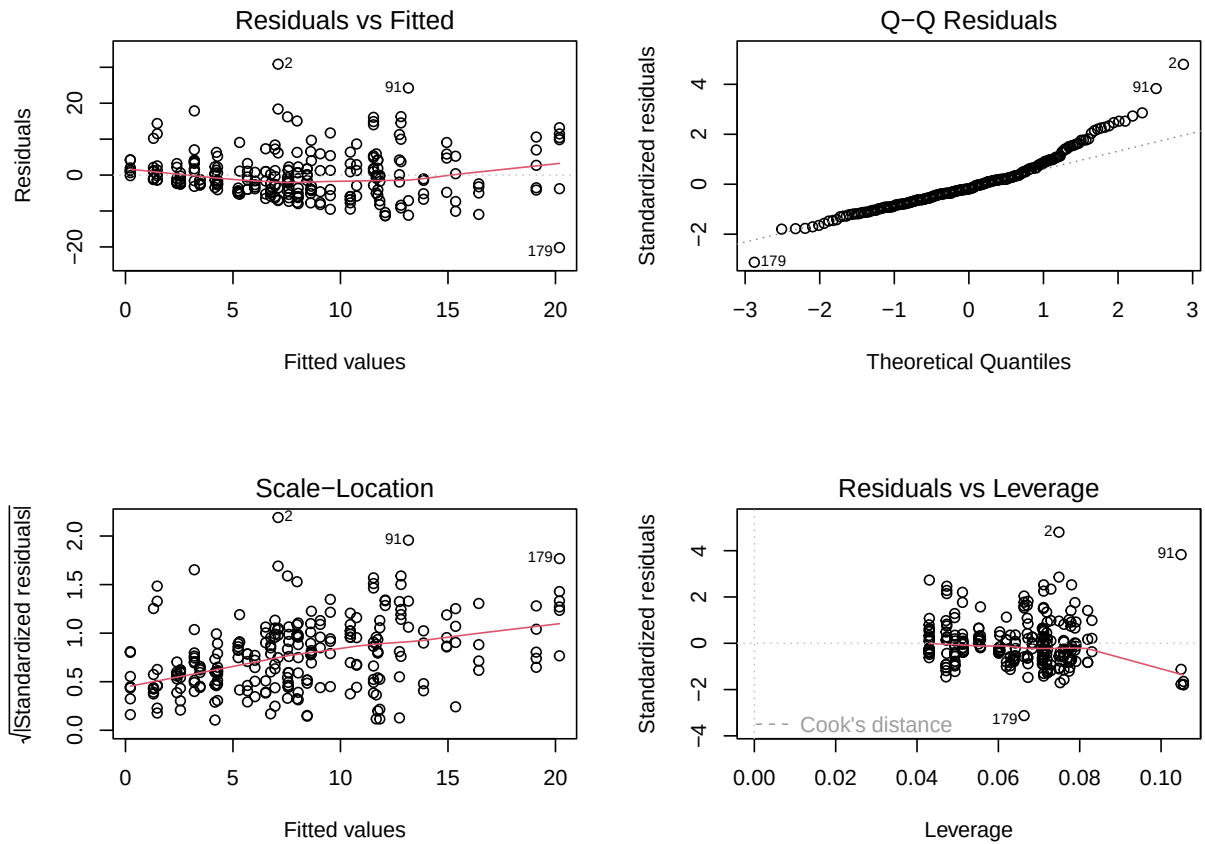
p_cas_sym <- make_panel(
  filter(confocal_data, marker == "Caspase", state == "Sym"),
  "Caspase - Sym",
  show_legend = TRUE,
  y_limit = ymax_cas
)

# Stats
fit_global <- lm(
  percentage ~ marker + state + heat * time + state * time,
  data = confocal_data
)

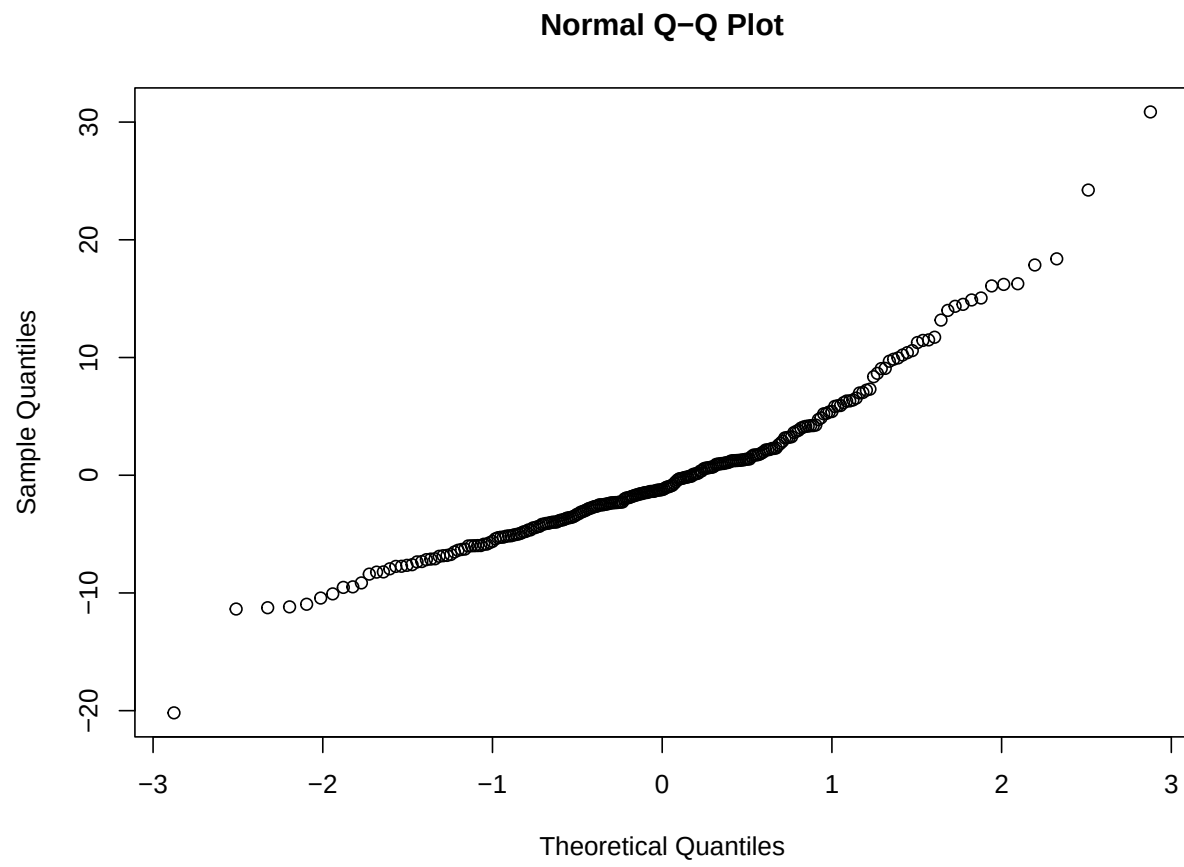
```

```
## Global model checks
```

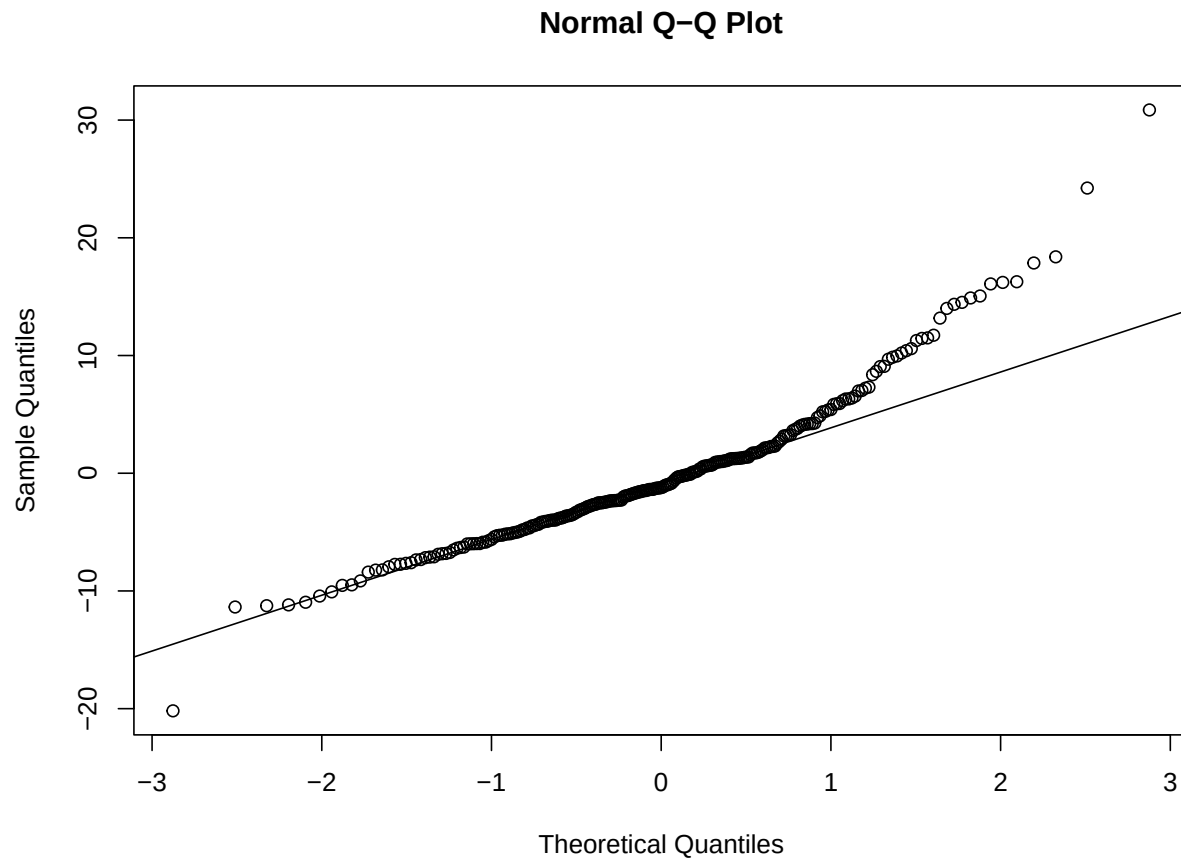
```
par(mfrow = c(2, 2))
plot(fit_global)
```



```
par(mfrow = c(1, 1))
qqnorm(residuals(fit_global))
```



```
qqline(residuals(fit_global))
```



```
shapiro.test(residuals(fit_global))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(fit_global)
## W = 0.93424, p-value = 4.264e-09
```

```
leveneTest(
  percentage ~ marker * state * heat * time,
  data = confocal_data
)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group 43  1.9537 0.001075 **
##      205
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Type II ANOVA

```
anova_global <- car::Anova(fit_global, type = 2)

anova_global_table <- as.data.frame(anova_global) %>%
  rownames_to_column("term") %>%
  mutate(
    across(where(is.numeric), ~ signif(.x, 3))
  )
anova_global_table
```

##	term	Sum Sq	Df	F value	Pr(>F)
## 1	marker	72.3	1	1.62	2.05e-01
## 2	state	1450.0	2	16.20	2.61e-07
## 3	heat	400.0	1	8.95	3.07e-03
## 4	time	385.0	3	2.87	3.71e-02
## 5	heat:time	537.0	3	4.00	8.33e-03
## 6	state:time	2460.0	5	11.00	1.55e-09
## 7	Residuals	10400.0	233	NA	NA

```
edu_apo <- confocal_data %>%
  filter(marker == "EdU", state == "Apo") %>%
  droplevels()

fit_edu_apo <- lm(percentage ~ heat * time, data = edu_apo)
anova_edu_apo <- car::Anova(fit_edu_apo, type = 2)
emm_edu_apo <- emmeans(fit_edu_apo, ~ heat | time)
tukey_edu_apo <- pairs(emm_edu_apo, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Apo")
letters_edu_apo <- cld(emm_edu_apo, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Apo")

edu_inoc <- confocal_data %>%
  filter(marker == "EdU", state == "Inoc") %>%
  droplevels()

fit_edu_inoc <- lm(percentage ~ heat * time, data = edu_inoc)
anova_edu_inoc <- car::Anova(fit_edu_inoc, type = 2)
emm_edu_inoc <- emmeans(fit_edu_inoc, ~ heat | time)
tukey_edu_inoc <- pairs(emm_edu_inoc, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Inoc")
letters_edu_inoc <- cld(emm_edu_inoc, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Inoc")

edu_sym <- confocal_data %>%
  filter(marker == "EdU", state == "Sym") %>%
  droplevels()

fit_edu_sym <- lm(percentage ~ heat * time, data = edu_sym)
```

```

anova_edu_sym <- car::Anova(fit_edu_sym, type = 2)
emm_edu_sym <- emmeans(fit_edu_sym, ~ heat | time)
tukey_edu_sym <- pairs(emm_edu_sym, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "EdU", state = "Sym")
letters_edu_sym <- cld(emm_edu_sym, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "EdU", state = "Sym")

cas_apo <- confocal_data %>%
  filter(marker == "Caspase", state == "Apo") %>%
  droplevels()

fit_cas_apo <- lm(percentage ~ heat * time, data = cas_apo)
anova_cas_apo <- car::Anova(fit_cas_apo, type = 2)
emm_cas_apo <- emmeans(fit_cas_apo, ~ heat | time)
tukey_cas_apo <- pairs(emm_cas_apo, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Apo")
letters_cas_apo <- cld(emm_cas_apo, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Apo")

cas_inoc <- confocal_data %>%
  filter(marker == "Caspase", state == "Inoc") %>%
  droplevels()

fit_cas_inoc <- lm(percentage ~ heat * time, data = cas_inoc)
anova_cas_inoc <- car::Anova(fit_cas_inoc, type = 2)
emm_cas_inoc <- emmeans(fit_cas_inoc, ~ heat | time)
tukey_cas_inoc <- pairs(emm_cas_inoc, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Inoc")
letters_cas_inoc <- cld(emm_cas_inoc, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Inoc")

cas_sym <- confocal_data %>%
  filter(marker == "Caspase", state == "Sym") %>%
  droplevels()

fit_cas_sym <- lm(percentage ~ heat * time, data = cas_sym)
anova_cas_sym <- car::Anova(fit_cas_sym, type = 2)
emm_cas_sym <- emmeans(fit_cas_sym, ~ heat | time)
tukey_cas_sym <- pairs(emm_cas_sym, adjust = "tukey") %>%
  as.data.frame() %>%
  mutate(marker = "Caspase", state = "Sym")
letters_cas_sym <- cld(emm_cas_sym, adjust = "tukey", Letters = letters) %>%
  as.data.frame() %>%
  mutate(.group = str_trim(.group), marker = "Caspase", state = "Sym")

# Supplemental table
anova_table <- bind_rows(

```

```

as.data.frame(anova_edu_apo) %>% rownames_to_column("term") %>% mutate(marker = "EdU", state = "EdU")
as.data.frame(anova_edu_inoc) %>% rownames_to_column("term") %>% mutate(marker = "EdU", state = "EdU")
as.data.frame(anova_edu_sym) %>% rownames_to_column("term") %>% mutate(marker = "EdU", state = "EdU")
as.data.frame(anova_cas_apo) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "Caspase")
as.data.frame(anova_cas_inoc) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "Caspase")
as.data.frame(anova_cas_sym) %>% rownames_to_column("term") %>% mutate(marker = "Caspase", state = "Caspase")
) %>%
  select(marker, state, term, everything())

supplemental_tukey_table <- bind_rows(
  tukey_edu_apo,
  tukey_edu_inoc,
  tukey_edu_sym,
  tukey_cas_apo,
  tukey_cas_inoc,
  tukey_cas_sym
) %>%
  select(marker, state, time, contrast, estimate, SE, df, t.ratio, p.value) %>%
  mutate(
    estimate = round(estimate, 2),
    SE = round(SE, 2),
    t.ratio = round(t.ratio, 2),
    p.value = signif(p.value, 3)
  ) %>%
  arrange(marker, state, time)

write_csv(anova_global_table, "~/Documents/GitHub/heating_lacerates_final/tables/TableS10-confocal_anova.csv")
write_csv(supplemental_tukey_table, "~/Documents/GitHub/heating_lacerates_final/tables/TableS11-confocal_anova.csv")

```